

Coursework-to-SOC Project Mapping

Coursework Requirement	How It Fits in SOC Simulation	Deliverable in Project
Incident Response Plan: Detection, containment, eradication, recovery, identify attack type (Malware, Phishing, Ransomware, DoS)	Build detection rules in SIEM (e.g., alert on suspicious login attempts or abnormal traffic). Run a simulated phishing email or malware-infected file event. Document containment (e.g., isolating endpoint VM), eradication, and recovery.	Incident Response Playbook for SOC lab + Simulation logs/screenshots of handling the event.
Comprehensive Security Policy: 3 key security rules/guidelines, CIA Triad, incident response	Develop lab policies like password policy , logging/monitoring requirements , and access controls . Show how these policies apply to your SOC lab environment and how they support Confidentiality, Integrity, Availability (CIA) .	Written SOC Security Policy + short demonstration (e.g., enforcing strong password policy on VMs).
Apply Encryption Techniques: AES encryption/decryption, hashing (MD5/SHA)	Integrate a script in your SOC project to encrypt/decrypt messages or hash log files to ensure integrity. For example, hash collected logs with MD5/SHA1 to detect tampering.	Python script in repo + demonstration logs showing encryption/decryption & file integrity checks.
Demonstrate Legal & Ethical Compliance: Include laws/regulations, 1 ethical consideration, explain how plan aligns	Map SOC actions to laws like HIPAA, GDPR , or Computer Fraud and Abuse Act . Discuss ethical duty to report breaches, avoid unnecessary exposure of personal data, etc.	Compliance Section inside Incident Response Plan document + reference to how SOC simulation avoids ethical/legal pitfalls.

Semester 3 Project Timeline with Dates

(September 2 – November 23, 2025)

Month 1 – Environment Setup & Security Foundations

- **Week 1 (Sep 2 – Sep 7)**
 - Install Parrot OS VM in VirtualBox
 - Install Wazuh SIEM + configure dashboard
 - Configure system logging (Parrot OS + macOS)
 - Draft **Security Policy Rule #1 – Password & Access Controls**
 - **Week 2 (Sep 8 – Sep 14)**
 - Deploy basic Wazuh agents (Parrot + macOS)
 - Configure log collection
 - Create **3 custom alert rules**
 - Draft **Security Policy Rule #2 – Logging & Monitoring**
 - **Week 3 (Sep 15 – Sep 21)**
 - Install & configure **Wireshark** (capture filters)
 - Install **Volatility** framework (memory analysis)
 - Draft **Security Policy Rule #3 – Incident Reporting Procedures**
 - **Week 4 (Sep 22 – Sep 28)**
 - Finalize **Comprehensive Security Policy Document** (CIA Triad + IR steps)
 - Document all environment setup with screenshots
 - Submit **Security Policy Coursework Deliverable** 
-

Month 2 – Detection, Incident Response & Containment

- **Week 5 (Sep 29 – Oct 5)**
 - Draft **Incident Response Plan** (detection → recovery)
 - Simulate **suspicious login attempt** → analyze logs in Wazuh
 - Build **event timeline + validate alerts**
 - **Week 6 (Oct 6 – Oct 12)**
 - Analyze **Parrot OS + macOS logs** in Wazuh
 - Classify **3 incidents** (severity matrix)
 - Simulate **phishing attack** → detect & contain
 - **Week 7 (Oct 13 – Oct 19)**
 - Configure **network isolation in Parrot OS** (firewall rules + interface config)
 - Implement **network segmentation in VirtualBox**
 - Expand **containment playbook**: isolation, traffic blocking, shutdown
 - **Week 8 (Oct 20 – Oct 26)**
 - Preserve evidence (filesystem, packet captures, memory dumps)
 - Finalize **Incident Response Plan with attack scenarios** (Malware, Phishing, Ransomware, DoS)
 - Submit **Incident Response Plan Coursework Deliverable** 
-

Month 3 – Evidence, Reporting & Post-Incident

- **Week 9 (Oct 27 – Nov 2)**
 - Perform **live data collection** (processes + network connections)

- Run **Volatility memory analysis** & document
 - Perform **disk acquisition** with imaging tools
 - Maintain **chain of custody documentation**
 - **Week 10 (Nov 3 – Nov 9)**
 - Build **IR Playbook**: commands + evidence procedures
 - Implement **incident tracking system** (spreadsheet or tool)
 - Create **incident report** for 1 event
 - Write **AES encryption/decryption script + hash logs (MD5/SHA)**
 - Submit **Encryption Techniques Coursework Deliverable** ✓
 - **Week 11 (Nov 10 – Nov 16)**
 - Conduct **system recovery** (VirtualBox + Parrot OS + network)
 - Perform **root cause analysis** (timeline + findings)
 - Add **Legal & Ethical Compliance Section** (2 laws + 1 ethical principle)
 - Submit **Legal & Ethical Compliance Deliverable** ✓
 - **Week 12 (Nov 17 – Nov 23)**
 - Create **Recovery Validation Checklist** (testing procedures)
 - Write **Process Improvement Recommendations** (lessons learned)
 - Finalize **all documentation & reports**
 - Submit **All Remaining Deliverables** ✓
-